

AN2DL - Second Challenge Report S.O.S.

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1 Introduction

For the second challenge, the objective was to **classify images** of human tissue to correctly label their molecular subtype. Many different architectures, training methods and data augmentation strategies were tested to find those that suited the task better. The ultimate performance indicator was the *F1 Score*:

$$F1\ Score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (1)$$

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \quad (2)$$

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives} \quad (3)$$

2 Problem Analysis

First, the datasets were inspected to have a clear idea of the task and its main challenges.

- The **training dataset** is made up of 691 pairs of images: the image proper and the mask.
 - the image proper is colored and represents the human tissue
 - the associated mask (b/w) identifies the regions of the image most likely to contain the diseased tissue

Each image belongs to one of 4 possible categorical labels.

- The **test dataset** is made up of 477 pairs of images, with the same structure as before

Some of the problems of the training set included:

- The presence of Shrek in some of the images, in a shape similar to that of the real tissue images. Shrek has a regular mask and is labeled as various different cells
- The presence of boogers added artificially to some of the images. The masks often points to the booger as a region of high importance.
- The variable dimension of the images. The shorter side is always 1024 pixels long, but the other one varies.
- The presence of some duplicates of the same tissue within the same image
- The presence of many areas devoid of information within the images
- The high resolution of the images
- The traces of marker on the images

3 Method

3.1 - Data Cleaning and preprocessing:

- **Image retrieval** to automatically remove images of Shrek (using an InceptionV3 pre-trained on ImageNet1k). The resulting similarity distances, both retrieving Shrek or retrieving tissues, were continuous: no distinct threshold was found able to separate Shrek from valid tissue. Therefore, the removal was carefully inspected to prevent discarding already scarce data.
- **Color based filtering** to remove duplicates containing boogers. Images were converted from RGB to HSV color space. An HSV range corresponding to the green tone of the booger was defined and the fraction of pixels in the image falling within this range was computed. Images exceeding a density threshold were discarded.
- **Patching** to preserve a high level of detail without having to deal with 1024×1024 images. First, the smallest rectangle containing the entire mask was found. If this rectangle was smaller than 255×255 (or whatever size was preferred for the specific architecture), padding was added up to that dimension. Then, masks were used to randomly sample a varying number of relevant pixels, and square patches of size 255×255 were extracted around the sampled pixels. Using the masks ensured that the processed data was mostly relevant.
- Images were **multiplied by their corresponding mask**, to filter just the relevant information. Due to a lack of performance improvement, the multiplication was later abandoned.

3.2 - Data Augmentation

- **Geometric transformations** (random horizontal/vertical flips, slight rotations) were applied dynamically during training to artificially increase the amount of data and to make the network invariant to the transformations applied. Strong rotations could not be applied to the original rectangular training images; they would go out of bounds with respect to a square, resulting in information loss.

- **Class balancing** was implemented by over-sampling minority classes to match the count of the most numerous class. This avoids bias toward the majority class

- **Automatic Augmentation**, as seen in [2] had poor results

3.3 - Model Architecture and Training

- Two CNNs were built from scratch, with the intent to process the masks along with the images:
 - one with **four channels**: three for RGB, one for the mask
 - one with **two separate pipelines**: one for images, one for masks in the FEN (inspired by [1]); features were subsequently concatenated and fed to the classifier as one vector

These CNNs from scratch were developed with **Layer normalization**, due to Batch normalization performing poorly on small batches.

- After that, **Transfer learning** was employed. Initially, the convolutional layers were frozen, and only the fully connected classification layer was trained. Later, **Fine Tuning** was performed, unfreezing various layers to adapt feature extraction to the task.
- A ResNet50 pretrained on human tissues with breast cancer (TCGA-BRCA) was employed, with very poor results
- **ResNet18** was preferred over ResNet50, which overfit due to its large depth and the small dataset size
- A **WideResNet50** pretrained on the Patch Camelyon (a dataset containing 96×96 histopathologic scans) was used.
- An attempt with a **DenseNet121** trained on ImageNet1k was made, with decent results
- An **EfficientNet_B3** trained on ImageNet1k showed comparatively good results with 300×300 patches
- An **EfficientNet_B4** trained on ImageNet1k showed the best results with 380×380 patches

Table 1: Performance summary

Model	Validation F1-score	Test F1-score
Four channels CNN	49.61%	31.12%
Double path CNN	33.61%	27.3%
WideResNet50	35.23%	29.48%
DenseNet121	37.15%	33.92%
ResNet18 transfer learning only	34%	32.41%
ResNet18 + fine tuning	43.89%	35.3%
EfficientNet_B4	45.63%	35.89%

- Various **optimizers** (Ranger, Adam, Lion) were tested

4 Experiments

The performance of some of the models built can be compared in Table 1.

5 Results

- The usage of a different optimizer with lower learning rate helped the network to converge.
- In more than one occasion the performance on the validation set did not follow a logical pattern. In the best scenario this could be related to an overfitting scenario. In the worst ones, they appeared to be completely random from the start.
- The best results with ResNets were obtained with just 18 layers, since deeper networks are more prone to overfitting.
- The best test performance has been obtained on the pretrained EfficientNet_B4, being designed to achieve higher performances with less parameters helped to reduce overfitting.
- Patching generally improved the results
- Imported networks trained on histological datasets did not necessarily perform better than networks trained on ImageNet. Fine Tuning was necessary whenever Transfer Learning was used.

6 Discussion

The networks did not develop necessarily to the F1 score's advantage. However, there are still some strong points in the presented approach:

- The **Color based filtering** was successful in removing all the boogers from the images
- The Patching strategy was proven to work, improving the F1 by enabling the networks to capture small features

7 Conclusions

Numerous different experiments were performed, with different architectures, parameters and data preprocessing techniques. Despite that, the variance in the test performance appears to be little more than fluctuations from the random initializations of the network's weights. It's strongly believed than some kind of error, perhaps not even strictly related to the network part, is acting as a bottleneck to our results, preventing us to gain noticeable performance improvements.

Once this error will be enlightened, numerous other paths will be available to explore, for example:

- Test-time augmentation
- Repeating the experiments that performed poorly, hoping to find better results, now hidden from our mysterious antagonist
- Implementing a transformers architecture

References

- [1] H. Alqasir, D. Muselet, and C. Ducottet. Mask-guided image classification with siamese net-

- works. In *Proceedings of the 15th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - Volume 5: VISAPP*, pages 536–543, Valetta, Malta, February 2020. SciTePress.
- [2] K. Faryna, J. van der Laak, and G. Litjens. Automatic data augmentation to improve generalization of deep learning in he stained histopathology. *Computers in Biology and Medicine*, 170:108018, 2024.